

① select speciality, count(*) as doctor_count,
Round (Avg(salary) as avg_salary, 2), sum(salary) as sum_salary
from doctors ~~where avg_salary > 120000~~ group by
speciality having avg_salary > 120000 order by
avg_salary desc;

② select d.doctor_name, d.speciality, s.doctor_name
from doctors d ~~left~~ left join doctors s
on d.supervisor_id = s.doctor_id

with cte as

③ select d.doctor_name, d.speciality, ^{ifnull} count(distinct(p.patient_id))
as patient_count from doctors d left join patients p on
d.doctor_id = p.doctor_id group by d.doctor_name

^{ifnull} select cte.doctor_name, cte.speciality, cte.patient_count,
~~ifnull~~ (sum(cost) as total_treatment_cost, 0) from
cte join treatments as t on cte.patient_id
= t.patient_id group by patient_id;

④ select patient_name, ward, age from patients
where patient_id in (select ^{distinct} patient_id) from
treatments where cost > 5000 order by
patient_name;

⑤ select d.doctor-name, d.speciality, d.salary,
 Rank (S.avg-salary, 2) as speciality-avg
 from doctors d join

(select speciality, Avg(salary) as Avg-salary
 from doctors) as S
 on d.speciality = S.speciality
 where d.salary > speciality-avg order by
 speciality, salary desc;

⑥ select d.doctor-name, d.speciality, d.salary,
 rank() over (partition by d.speciality
 as sal-rank order by salary desc)
 from doctors d
 order by speciality, sal-rank;

⑦ select p.patient-name, t.treatment, t.treat-date,
 t.cost, Sum(t.cost) over (partition by p.patient-
 name order by t.treat-date) as
 running cost
 from treatments t
 join patients p on t.patient-id = p.patient-id
 order by p.patient-name, t.treat-date;